

ASSIGNMENT #05

Course: CHE 341: CHEMICAL PROCESS CONTROL
Term: Spring 2026
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Due Date: Monday, March 30th in class

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Prepare a **typed** report of your assignment.
 - Include this cover page with your name in the report.
 - Upload your report as a PDF file by the Due Date and Time(**6:00PM**) on the UIC Blackboard.
 - Bring a hard copy of your report as a PDF file by the Due Date and Time(**Beginning of the Lecture**) in class.

 - Assignments that are not typed will not be accepted.
 - Late Assignments will not be accepted without prior approval.
 - You may use the Internet as well as ChatGPT or other public tools as long as you place a written statement about their use in your assignment report
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Problem A: Empirical Model Building

Construct the process transfer function from the multiple step test commercial data contained in the accompanying Excel file: **Assignment-05.xlsx**

To construct the process transfer function, the data from the provided excel file was utilized. In the second sheet titled “Transfer Function,” the PV(t) and MV(t) datasets were used to model the system dynamics. A separate time column was created, dedicated quantifying the time into two second intervals.

Two deviation columns were created for the Manipulated and Process Variables to simulate the model. Which are expressed as:

$$\begin{aligned}\Delta MV(t) &= MV(t) - MV_{initial} \\ \Delta PV(t) &= PV(t) - PV_{initial}\end{aligned}$$

Next, initial guesses were made for the transfer function parameters: K_p , θ , and τ . The initial guesses were $K_p = 1$, $\theta = 2$, and $\tau = 10$.

Another column was created for Predicted PV and a numerical integration formula was utilized for the second row and onwards.

The equation used is:

$$Predicted\ PV_{New} = Predicted\ PV_{Old} + \left(\frac{\Delta t}{\tau}\right) \times (K_p \cdot \Delta MV_{Delayed} - Predicted\ PV_{Old})$$

$\Delta MV_{Delayed}$ is to simulate the physical lag in the system, representing the delay between input change and the physical response. The index function was used in Excel with a five-row delay.

Afterwards, an Error Squared Column was created. Each row is the difference between the actual and predicted process variables squared. In a separate cell, the values had their summations taken.

Excel Solver was then used to find the new values of K_p , θ , and τ for the Process Transfer Function equation. The values were found by setting Error Squared Column as the objective, changing variable cells were K_p , θ , and τ , and the constraints were defined as $\tau > 0$ and $\theta \geq 0$.

After running, the new values were found to be:

$$K_p = 3.23$$

$$\theta = 2$$

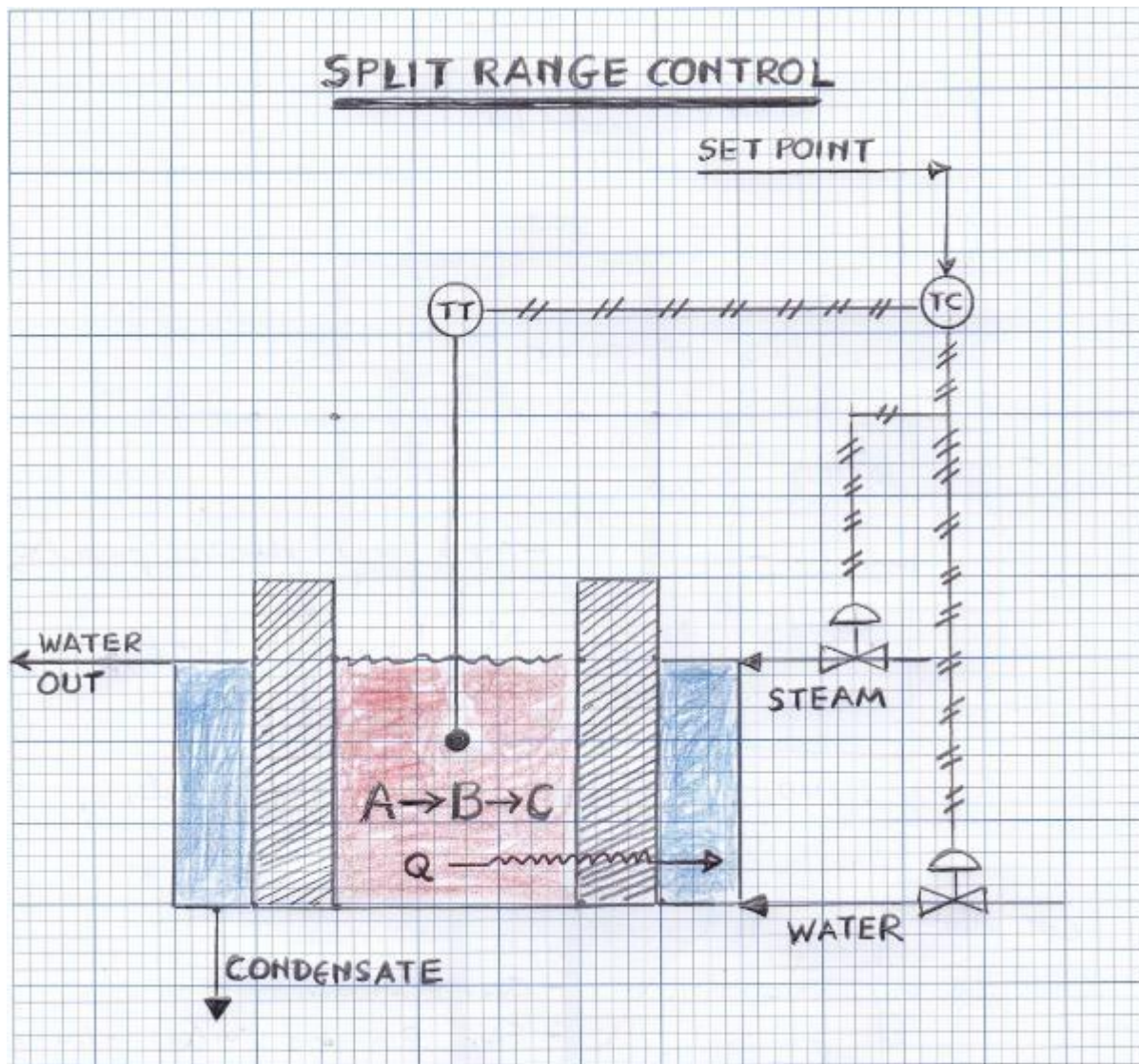
$$\tau = 30.24$$

The values can be inserted into the First Order Plus Time Delay equation to yield the final process transfer function:

$$G(s) = \frac{K_p e^{-\theta s}}{\tau s + 1}$$
$$G(s) = \frac{3.23 e^{-2s}}{30.24s + 1}$$

Problem B: Process Control Loops and Measurement Devices for Split Range Control

A process diagram for the control of a batch reactor is shown in the Figure below:



The size of the batch reactor (red color), the metal wall thickness (black pattern) and the jacket thickness (blue color) are not drawn to scale. The objective of this assignment is to control the production of a specialty chemical compound that takes place in the batch reactor. The specialty chemical is produced by the following reactions, with the adjacent kinetic data:



ΔE	ΔE
$k_1 = k_{10} \exp\left[-\frac{\Delta E_1}{RT}\right] C_A$	$k_2 = k_{20} \exp\left[-\frac{\Delta E_2}{RT}\right] C_B$
$E_1 = 15,000 \text{ Btu/lbmol}$	$E_2 = 20,000 \text{ Btu/lbmol}$
$k_{10} = 729.55 \text{ min}^{-1}$	$k_{20} = 6567.6 \text{ min}^{-1}$
$\Delta H_1 = -40,000 \text{ Btu/lbmol}$	$\Delta H_2 = -50,000 \text{ Btu/lbmol}$

Batch processes offer some very distinct and challenging control problems. In a typical scenario (like the one in this assignment) the reactants are charged in the reactor which also may contain a

solvent and the reactor temperature is raised by using a temperature controller that controls the flow rate of steam in the jacket of the reactor (see Figure above). Once the temperature reaches the desired point, the controller shuts down the steam valve. At this point the chemical reactions are initiated either by charging a catalyst or simply because of the high temperature. If the chemical reactions are exothermic the temperature will begin to rise. This can be detrimental to the product as well as the catalyst. To avoid a large temperature rise, the temperature controller will now open the water valve and start manipulating the water flow rate, to keep the reactor temperature at a specified set point. This is known as **split range control**.

Set points for the temperature are pre-calculated by performing optimization studies which are designed to maximize the production of the specialty chemical. Most of the time the set point varies with time and this variation is calculated by the nonlinear optimizers. In our example the set point has been pre-calculated and is given below. To control the temperature at its set point you will design a proportional controller. Your objective would be to make as much product as possible, keeping the process stable and avoiding large oscillations in the temperature and the water flow rate. Large water flow oscillations result in wear and tear of the valve. The following data are available for you to conduct your studies:

Process Variable	Reactor	Metal Wall	Reactor Jacket	Water
Initial Temperature, F °	80	80		80
Initial Concentrations, lbmol/ft ³	$C_{Ao} = 0.80$ $C_{Bo} = 0.00$ $C_{Co} = 0.00$			
Volume, ft ³	42.5	9.42	18.83	
Heat Transfer Area, ft ²	56.5 → to metal wall	56.5 → to jacket		
Heat Transfer Coeff., Btu/(hr ft F) ² °	160 → to metal wall	400 → to water 1000 → to steam		
Heat Capacity, Btu/(lb F)°	1.00	0.12		1.0
Density, lb/ft ³	50.0	512.0		62.3

Cooling water is available at 80 °F. The temperature sensor(thermocouple) has a time constant equal to $\tau_s=1$ min. The temperature transmitter sends a pneumatic signal to the temperature controller. The pneumatic signal range is [3 psi – 15 psi]. The reactor temperature range is [50 °F - 250 °F]. The water valve is linear and has been sized with $C_v=100$ gpm (psi)^{-0.5}[fully open]. The pressure drop across the valve is constant and equal to 20 psi. The valve is fully closed at a pressure signal of 9 psi and fully open at a pressure signal of 3 psi. The steam valve is also linear and has been sized to give a valve coefficient $C_v=112$ lb/min (psi)^{-0.5}[fully open]. The pressure drop across the valve is constant and equal to 0.2 psi. The valve is fully closed at a pressure signal of 9 psi and fully open at a pressure signal of 15 psi.

The controller has a bias of 7 psi and its output signal is given by

1 t

$$CO(t) = 7[\text{psi}] + K_c (SP - TT) + \frac{1}{\Delta t} \int_0^{\Delta t} (SP - TT) dt$$

where **TT** is the temperature transmitter signal in psi and **SP** the set point in psi. The set point has been calculated via optimization and it is given by:

$$SP = 12[\text{psi}]$$

The total time of the batch process is **60 min**. Use the 5th order Runge-Kutta method with Total Number of Solution Points=200 and Number of Steps between Solution Points=1

THE FOLLOWING TASKS ARE REQUIRED OF EACH TEAM

1. Prepare the dynamic model for this batch process. You may neglect the dynamics of the steam (but not of water) in the jacket. The reactor metal wall must also be included in the energy balances. Include a description of all symbols that you have used. The mass and energy balances must be written clearly for the three phases of the specialty chemical production, i.e.,

These are the symbols that will be used for the dynamic model and their definitions:

V_r, V_j : Volume of the reactor mixture and jacket

ρ_w : Density of cooling water

m_r, m_m, m_j : Mass of the reactor mixture, metal wall, and cooling water in the jacket

C_{pr}, C_{pm}, C_{pw} : Heat capacities of the reactor mixture, metal wall, and cooling water

C_A, C_B, C_C : Concentrations of the chemical components A, B, and C

T, T_m, T_j : Temperatures of the reactor mixture, metal wall, and jacket water

T_s : Temperature of the saturated steam at 35 psia

T_{win} : Inlet temperature of the cooling water

A_i, A_o : Heat transfer areas from the reactor fluid to the metal wall and from the metal wall to the jacket

h_i, h_s, h_w : Heat transfer coefficients for the reactor fluid, steam, and water

R_1, R_2 : Reaction rates for the conversion of A to B and B to C

k_{10}, k_{20} : Pre-exponential factors

E_1, E_2 : Activation energies

$\Delta H_1, \Delta H_2$: Enthalpies of reaction

F_w : Mass flow rate of the cooling water

The specialty chemical is produced in the batch reactor is defined by the reaction $A \xrightarrow{k_1} B \xrightarrow{k_2} C$ and occurs in two steps. The reaction rates are expressed by the given equations:

$$R_1 = k_{10} \exp\left\{-\frac{E_1}{RT}\right\} C_A$$

$$R_2 = k_{20} \exp\left\{-\frac{E_2}{RT}\right\} C_B$$

It is assumed that there will be constant liquid volume in the batch reactor; as such, mass balances for the chemical components will be applied universally across all three phases.

They are expressed as:

$$-R_1 = \frac{dC_A}{dt}$$

$$R_1 - R_2 = \frac{dC_B}{dt}$$

$$R_2 = \frac{dC_C}{dt}$$

- a. Heating phase using dry saturated steam from a **35 psia** steam header flowing through the reactor jacket

To formulate the energy balance for reactor fluid during the heating phase, the First Law of Thermodynamics is applied to the batch reactor. Due to the absence of mass entering or exiting the process and constant volume, focus is placed on the rate of internal energy.

The energy balance for a non-flow batch system can be defined as:

$$\text{Rate of Accumulation} = \text{Rate of Heat Transfer} + \text{Rate of Heat Generation by Reaction}$$

Assuming constant heat capacity, the rate of energy is expressed as:

$$\text{Accumulation} = m_r C_{pr} \frac{dT}{dt}$$

Newton's Law of Cooling will be utilized to determine the heat capacity, where the temperature difference is between the metal wall (T_m) and reactor mixer (T):

$$Q = h_i A_i (T_m - T)$$

The specialty chemical is produced by two successive exothermic reactions; as such, the heat generated by the process is expressed as:

$$\text{Heat Generation} = (-\Delta H_1)R_1V_r + (-\Delta H_2)R_2V_r$$

The terms can be combined to formulate the final differential equation for the reactor fluid temperature:

$$m_r C_{pr} \frac{dT}{dt} = h_i A_i (T_m - T) + (-\Delta H_1)R_1V_r + (-\Delta H_2)R_2V_r$$

A similar process can be applied for the metal wall with the exception of the previous energy generation term which only applies to the reactor fluid.

$$\text{Accumilation} = m_m C_{pm} \frac{dT_m}{dt}$$

$$\text{Rate of heat in} = h_s A_o (T_s - T_m)$$

$$\text{Rate of heat out} = h_i A_i (T_m - T)$$

Combined, the terms form the final equation:

$$m_m C_{pm} \frac{dT_m}{dt} = h_s A_o (T_s - T_m) - h_i A_i (T_m - T)$$

b. Cooling phase using cooling water filling the reactor jacket

The energy balance for the reactor fluid will remain the same as it does not come into contact with either the cooling water or steam.

$$m_r C_{pr} \frac{dT}{dt} = h_i A_i (T_m - T) + (-\Delta H_1)R_1V_r + (-\Delta H_2)R_2V_r$$

The metal wall, however, will face a slight variation due to the absence of steam and the introduction of cooling water. Heat is no longer supplied by the steam header; instead, energy is transferred from the metal wall to the cooling water in the jacket.

The new rates of heat are expressed as:

$$\text{Rate of heat in} = h_i A_i (T_m - T)$$

$$\text{Rate of heat out} = h_w A_o (T_m - T_j)$$

The new balance equation is:

$$m_m C_{pm} \frac{dT_m}{dt} = -h_i A_i (T_m - T) - h_w A_o (T_m - T_j)$$

Before the energy balance equation for the reactor jacket can be formulated, the mass balance must be defined. This is due to the jacket transitioning from an empty vessel to a filled one; the mass is constantly accumulating. As such, both the mass and balance equations must be accommodated for simultaneously.

Water is entering the jacket at a constant mass flow rate which can be defined as F_w . Since there is no water exiting the system at this stage, the rate of change of the mass of water can be expressed as:

$$\frac{dm_j}{dt} = F_w$$

$$\text{Rate of Accumulation} = \text{Energy In (via flow)} + \text{Heat Transfer (from wall)}$$

The total energy in the jacket is defined as:

$$U = m_j C_{pw} T_j$$

Therefore, the rate is expressed as:

$$\frac{d(m_j C_{pw} T_j)}{dt} = F_w C_{pw} T_{win} + h_w A_o (T_m - T_j)$$

The product rule will be applied to expand the left side of the equation and substituted into the total energy balance:

$$\begin{aligned} \frac{d(m_j C_{pw} T_j)}{dt} &= C_{pw} \left(m_j \frac{dT_j}{dt} + T_j \frac{dm_j}{dt} \right) \\ C_{pw} m_j \frac{dT_j}{dt} + C_{pw} T_j \frac{dm_j}{dt} &= F_w C_{pw} T_{win} + h_w A_o (T_m - T_j) \end{aligned}$$

The equation will now be simplified to reach the final equation.

Substituting F_w into $\frac{dm_j}{dt}$:

$$C_{pw}m_j \frac{dT_j}{dt} + C_{pw}T_j F_w = F_w C_{pw}T_{win} + h_w A_o (T_m - T_j)$$

Subtracting F_w to isolate $C_{pw}m_j \frac{dT_j}{dt}$:

$$C_{pw}m_j \frac{dT_j}{dt} = F_w C_{pw}T_{win} - F_w C_{pw}T_j + h_w A_o (T_m - T_j)$$

Factoring out $F_w C_{pw}$:

$$m_j C_{pw} \frac{dT_j}{dt} = F_w C_{pw} (T_{win} - T_j) + h_w A_o (T_m - T_j)$$

c. Operating with jacket full of water.

For the operating jacket full of water, the energy balances for both the reactor fluid and metal wall will remain unchanged from the cooling phase. For the former, the fluid only contacts the inner metal wall; therefore, heat transfer and temperatures remain the same. Additionally, for the metal wall, the water is already touching water in both of the phases, so heat transfer definitions will remain identical.

Since the jacket is now full, the mass of water is constant and determined by two factors: volume (V_j) and density of water (ρ_w).

$$m_j = V_j \rho_w$$

Mass is also no longer accumulating, meaning the mass flowrate in is equal to the mass flowrate out. The rate of change ($\frac{dm_j}{dt}$) will be equal to zero.

$$\frac{dm_j}{dt} = 0$$

The new energy balance terms are:

$$\begin{aligned} \text{Accumulation} &= \text{Energy In (Flow)} - \text{Energy Out (Flow)} + \text{Heat Transfer (from Wall)} \\ &= m_j C_{pw} \frac{dT_j}{dt} \end{aligned}$$

$$\text{Energy In} = F_w C_{pw} T_{win}$$

$$\text{Energy Out} = F_w C_{pw} T_j$$

$$\text{Heat Transfer} = h_w A_o (T_m - T_j)$$

The terms are combined to form:

$$m_j C_{pw} \frac{dT_j}{dt} = F_w C_{pw} T_{win} - F_w C_{pw} T_j + h_w A_o (T_m - T_j)$$

It is then simplified via factoring out $F_w C_{pw}$ to net the final equation:

$$m_j C_{pw} \frac{dT_j}{dt} = F_w C_{pw} (T_{win} - T_j) + h_w A_o (T_m - T_j)$$

2. Solve the differential equations using the **CHE Calculator** and prepare plots of the reactor temperature for controller gains $K_c = 2, 4, 8, 10$ and $\tau_c = 1, 2, 4, 8, 10$. Give a brief description of what you are observing as you increase the controller gain. Include an Appendix in your report with the Visual Basic code of your model and the excel worksheet with your data and the plots.

I apologize; I was unable to construct a functioning code through CHE Calculator's VBA function.

3. Investigate the use of PI controller to eliminate the offset and propose a value for the controller gain and the controller rest. For these values create a plot for the reactor temperature as a function of time.

Once more I apologize. I am unable to solve this problem.